Custom Sentinel Alert Rules

By

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IOC-Based Detection Engineering

TOTAL RULES	CRITICAL	HIGH	TASKS COVERED
8	6	2	1, 2, 3

1. Overview

This document details the eight custom analytics rules created in Microsoft Sentinel as part of the CyBlack Security Operations Sprint. Each rule is derived from Indicators of Compromise (IOCs) and behavioural patterns identified during the active investigations in Tasks 1 (Phishing), 2 (Account Compromise), and 3 (WannaCry Malware Analysis).

The rules provide ongoing detection capability for threat actor infrastructure and attack techniques observed during the sprint, ensuring recurrence of the same campaigns or IOCs triggers an automated alert in the SOC queue. All rules follow scheduled query format in Microsoft Sentinel and are mapped to the MITRE ATT&CK framework.

2. Consolidated IOC List

2.1 Task 2 - Phishing Campaign IOCs

Type	Value	Description
Email Address	alfredegov@gmail.com	Threat actor - confirmed sender of Emails 4 and 5
Email Address	zelatcol@gmail.com	NjRAT delivery - Email 2 to Michelle Jackson
Email Address	projectdept@kanzalshamsprojectmgt.com	Phishing sender - Email 3 to Nike Fury
Domain	wtools.io	NjRAT malware distribution platform
Domain	kanzalshamsprojectmgt.com	Phishing domain - 4/94 VT detections
Domain	updetadobe.blogspot.com	Fake Adobe staging page - malware flagged
Domain	theannoyingsite.com	Browser attack toolkit - 13/94 VT detections
Domain	azxmarket.com	WannaCry delivery sender infrastructure
Domain	duchessgarden.sn	WannaCry delivery source domain
URL	https://wtools.io/paste-code/bOs4	NjRAT payload URL
URL	http://vmi1159541.contaboserver.net/payload.exe	Direct malware download
IP Address	209.145.51.44	Contabo C2 server IP

2.2 Task 2 - Account Compromise IOCs

Type	Value	Description
IP Address	79.124.60.6	Sofia, Bulgaria - VenomDC bulletproof hosting
IP Address	150.158.77.170	Shanghai, China - Tencent Cloud - HTTP bruteforce confirmed
IP Address	124.120.140.202	Bangkok, Thailand - Azure CLI attack attempt
File Hash (SHA256)	288b53495ef3fd237fa4640756c096bc8daaa6c6ff4942f6f792b29038ec259e	caffeine.exe - mouse jiggler policy violation
File	updated_payroll.xlsx	Sensitive payroll data downloaded by attacker
File	vulnerability_report_June2023.docx	Security vulnerability report accessed by attacker
Request ID	Sdfvasdfv-9423-fsafda-asd-sdafsfsadf	Malformed UUID - automated tooling indicator

2.3 Task 3 - WannaCry IOCs

Type	Value	Description
SHA-256	ed01ebfbc9eb5bbea545af4d01bf5f10716 61840480439c6e5babe8e080e41aa	diskpart.exe - WannaCry primary binary
MD5	84c82835a5d21bbcf75a61706d8ab549	diskpart.exe
SHA-1	5ff465afaabcbf0150d1a3ab2c2e74f3a442 6467	diskpart.exe
MD5	4fef5e34143e646dbf9907c4374276f5	taskdl.exe - encryption worker
Filename	@WanaDecryptor@.exe	WannaCry ransom UI
Filename	@Please_Read_Me@.txt	WannaCry ransom note
File Extension	*.WNCRY	Encrypted victim file extension
Registry Key	HKLM\SOFTWARE\WannaCrypt	WannaCry infection indicator
Startup File	~SDCE9B.tmp	Startup folder persistence
Tor Onion	57g7spgrzlojinas.onion	WannaCry C2 payment server
Domain	www.iuqerfsodp9ifjaposdfjhgosurijfaewr wergwea.com	WannaCry kill switch - DO NOT BLOCK

Email Sender	c6d90d50-df87-11ee-a2fc-525400450766@azxmarket.com	WannaCry delivery sender

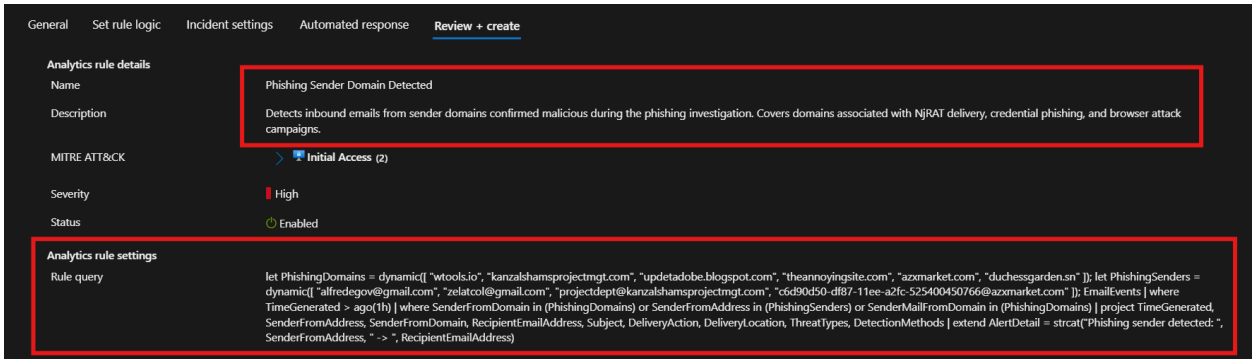
3. Alert Rules Summary

#	Rule Name	Severity	Schedule	Lookback	Source Task
1	Phishing Sender Domain Detected	HIGH	1 Hour	1 Hour	Task 2
2	Malicious Domain or C2 Contact Detected	CRITICAL	30 Min	30 Min	Task 2
3	Sign-In from Malicious or Foreign IP	CRITICAL	15 Min	15 Min	Task 3
4	WannaCry File Hash / Process Detection	CRITICAL	15 Min	15 Min	Task 4
5	WannaCry Registry Persistence / Shadow Copy Deletion	CRITICAL	15 Min	15 Min	Task 4
6	WannaCry C2 / Tor Communication / SMB Lateral Movement	CRITICAL	15 Min	15 Min	Task 4
7	Account Compromise Behavioural Pattern	CRITICAL	15 Min	2 Hours	Task 3
8	Coordinated Phishing Campaign Against Multiple Staff	HIGH	30 Min	1 Hour	Task 2

4. Rule Definitions

RULE 1	Phishing Sender Domain Detected	HIGH
Field	**Value**	
Description	Detects inbound emails from sender domains and addresses confirmed malicious during Task 2. Covers NjRAT delivery, credential phishing, browser attack, and WannaCry delivery infrastructure.	
MITRE Tactic	Initial Access	
MITRE Technique	T1566 - Phishing | T1566.001 - Spearphishing Attachment	
Run Frequency	Every 1 Hour	
Lookback Window	1 Hour	
Alert Threshold	Greater than 0 results	
Entity Mapping	Account (RecipientEmailAddress), Account (SenderFromAddress)	
Grouping	By SenderFromDomain within 1 Hour	

KQL Query



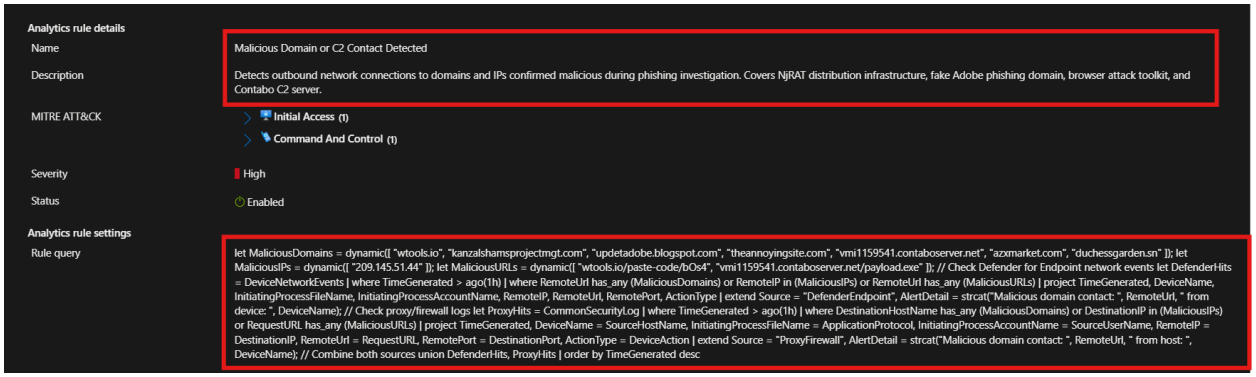
RULE 1 - Sentinel Analytics rule saved, showing High severity and Enabled status

```
let PhishingDomains = dynamic(["wtools.io", "kanzalshamsprojectmgt.com", "updetadobe.blogspot.com", "theannoyingsite.com",
"azxmarket.com", "duchessgarden.sn"]);
let PhishingSenders = dynamic(["alfredegov@gmail.com", "zelatcol@gmail.com", "projectdept@kanzalshamsprojectmgt.com",
"c6d90d50-df87-11ee-a2fc-525400450766@azxmarket.com"]);
EmailEvents
| where TimeGenerated > ago(1h)
| where SenderFromDomain in (PhishingDomains) or SenderFromAddress in (PhshingSenders) or SenderMailFromDomain in
(PhishingDomains)
| project TimeGenerated, SenderFromAddress, SenderFromDomain, RecipientEmailAddress, Subject, DeliveryAction, DeliveryLocation,
ThreatTypes, DetectionMethods
| extend AlertDetail = strcat("Phishing sender detected: ", SenderFromAddress, " -> ", RecipientEmailAddress)
```

RULE 1 KQL

RULE 2	Malicious Domain or C2 Contact Detected	CRITICAL
Field	**Value**	
Description	Detects outbound network connections to domains and IPs confirmed malicious during Task 2. Queries both Defender for Endpoint and proxy/firewall logs simultaneously.	
MITRE Tactic	Command and Control | Initial Access	
MITRE Technique	T1071 - Application Layer Protocol | T1566 - Phishing	
Run Frequency	Every 30 Minutes	
Lookback Window	30 Minutes	
Alert Threshold	Greater than 0 results	
Entity Mapping	Host (DeviceName), IP (RemoteIP), URL (RemoteUrl)	
Grouping	By DeviceName within 30 Minutes	

KQL Query



Analytics rule details

Name: Malicious Domain or C2 Contact Detected

Description: Detects outbound network connections to domains and IPs confirmed malicious during phishing investigation. Covers NjRAT distribution infrastructure, fake Adobe phishing domain, browser attack toolkit, and Contabo C2 server.

MITRE ATT&CK: Initial Access (1); Command And Control (1)

Severity: High

Status: Enabled

Analytics rule settings

Rule query:

```
let MaliciousDomains = dynamic([ "wtools.io", "kanzalshamsprojectmgt.com", "updetadobe.blogspot.com", "theannoyingsite.com", "vmi1159541.contaboserver.net", "azxmarket.com", "duchessgarden.sn" ]); let MaliciousIPs = dynamic([ "209.145.51.44" ]); let MaliciousURLs = dynamic([ "wtools.io/paste-code/bOs4", "vmi1159541.contaboserver.net/payload.exe" ]); // Check Defender for Endpoint network events let DefenderHits = DeviceNetworkEvents | where TimeGenerated > ago(1h) | where RemoteUrl has_any (MaliciousDomains) or RemoteIP in (MaliciousIPs) or RemoteUrl has_any (MaliciousURLs) | project TimeGenerated, DeviceName, InitiatingProcessFileName, InitiatingProcessAccountName, RemoteIP, RemoteUrl, RemotePort, ActionType | extend Source = "DefenderEndpoint", AlertDetail = strcat("Malicious domain contact: ", RemoteUrl, " from device: ", DeviceName); // Check proxy/firewall logs let ProxyHits = CommonSecurityLog | where TimeGenerated > ago(1h) | where DestinationHostName has_any (MaliciousDomains) or DestinationIP in (MaliciousIPs) or RequestURL has_any (MaliciousURLs) | project TimeGenerated, DeviceName = SourceHostName, InitiatingProcessFileName = ApplicationProtocol, InitiatingProcessAccountName = SourceUserName, RemoteIP = DestinationIP, RemoteUrl = RequestURL, RemotePort = DestinationPort, ActionType = DeviceAction | extend Source = "ProxyFirewall", AlertDetail = strcat("Malicious domain contact: ", RemoteUrl, " from host: ", DeviceName); // Combine both sources union DefenderHits, ProxyHits | order by TimeGenerated desc
```

RULE 2 - Sentinel Analytics rule saved, showing Critical severity and Enabled status

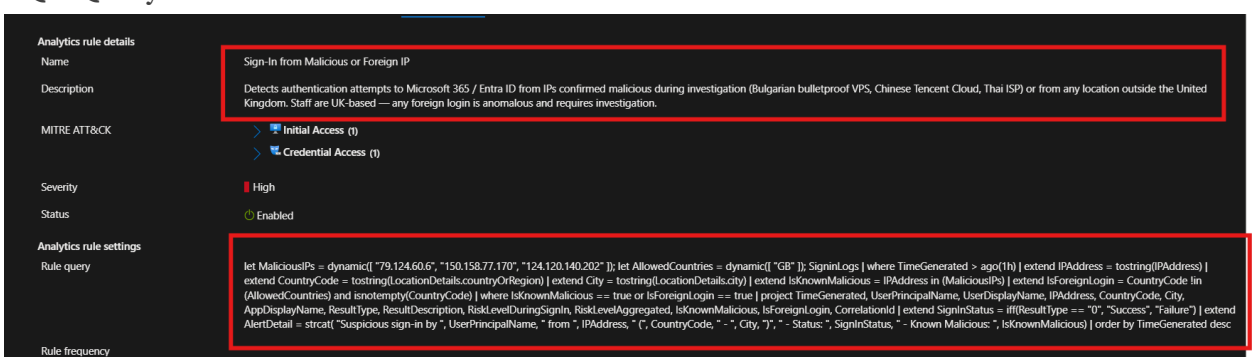
```
let MaliciousDomains = dynamic(["wtools.io", "kanzalshamsprojectmgt.com", "updetadobe.blogspot.com", "theannoyingsite.com",
"vmi1159541.contaboserver.net", "azxmarket.com", "duchessgarden.sn"]);
let MaliciousIPs = dynamic(["209.145.51.44"]);
let DefenderHits = DeviceNetworkEvents
| where TimeGenerated > ago(30m)
| where RemoteUrl has_any (MaliciousDomains) or RemoteIP in (MaliciousIPs)
| project TimeGenerated, DeviceName, InitiatingProcessFileName, InitiatingProcessAccountName, RemoteIP, RemoteUrl, RemotePort,
ActionType
| extend Source = "DefenderEndpoint", AlertDetail = strcat("Malicious domain contact: ", RemoteUrl, " from device: ", DeviceName);
let ProxyHits = CommonSecurityLog
| where TimeGenerated > ago(30m)
| where DestinationHostName has_any (MaliciousDomains) or DestinationIP in (MaliciousIPs)
| project TimeGenerated, DeviceName = SourceHostName, InitiatingProcessFileName = ApplicationProtocol, InitiatingProcessAccountName =
SourceUserName, RemoteIP = DestinationIP, RemoteUrl = RequestURL, RemotePort = DestinationPort, ActionType = DeviceAction
| extend Source = "ProxyFirewall", AlertDetail = strcat("Malicious domain contact: ", RemoteUrl, " from host: ", DeviceName);
union DefenderHits, ProxyHits | order by TimeGenerated desc
```

RULE 2 KQL

RULE 3	Sign-In from Malicious or Foreign IP	CRITICAL

Field	Value
Description	Detects Entra ID authentication from the three confirmed malicious IPs from Task 3, or any IP geolocated outside the United Kingdom. Staff are UK-based - any foreign login is anomalous.
MITRE Tactic	Initial Access | Credential Access
MITRE Technique	T1078 - Valid Accounts | T1110 - Brute Force
Run Frequency	Every 15 Minutes
Lookback Window	15 Minutes
Alert Threshold	Greater than 0 results
Entity Mapping	Account (UserPrincipalName), IP (IPAddress)
Grouping	By UserPrincipalName within 1 Hour

KQL Query

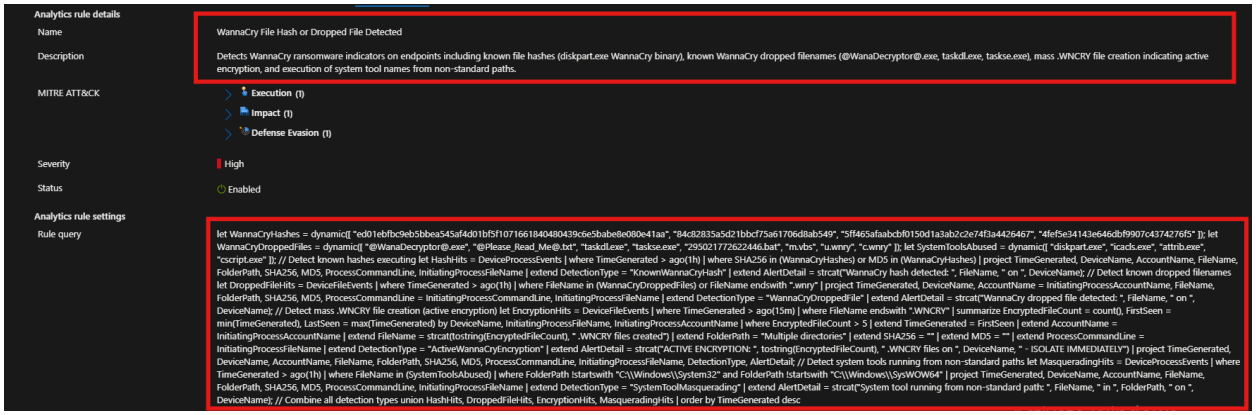


RULE 3 - Sentinel Analytics rule saved, showing Critical severity and Enabled status

```
let MaliciousIPs = dynamic(["79.124.60.6", "150.158.77.170", "124.120.140.202"]);
let AllowedCountries = dynamic(["GB"]);
SigninLogs
| where TimeGenerated > ago(15m)
| extend CountryCode = tostring(LocationDetails.countryOrRegion), City = tostring(LocationDetails.city)
| extend IsKnownMalicious = IPAddress in (MaliciousIPs)
| extend IsForeignLogin = CountryCode !in (AllowedCountries) and isnotempty(CountryCode)
| where IsKnownMalicious == true or IsForeignLogin == true
| project TimeGenerated, UserPrincipalName, UserDisplayName, IPAddress, CountryCode, City, AppDisplayName, ResultType,
ResultDescription, RiskLevelDuringSignIn, IsKnownMalicious, IsForeignLogin
| extend SignInStatus = iff(ResultType == "0", "Success", "Failure")
| extend AlertDetail = strcat("Suspicious sign-in by ", UserPrincipalName, " from ", IPAddress, " (", CountryCode, " - ", City, ") Status: ",
SignInStatus, " Known Malicious: ", IsKnownMalicious)
```

RULE 4	WannaCry File Hash / Process Detection	CRITICAL
Field	**Value**	
Description	Detects WannaCry ransomware on endpoints via known file hashes, dropped filenames, mass .WNCRY file creation indicating active encryption, and system tools executing from non-standard paths.	
MITRE Tactic	Execution | Impact | Defense Evasion	
MITRE Technique	T1204 - User Execution | T1486 - Data Encrypted for Impact | T1036 - Masquerading	
Run Frequency	Every 15 Minutes	
Lookback Window	15 Minutes	
Alert Threshold	Greater than 0 results	
Entity Mapping	Host (DeviceName), Account (AccountName), File (FileName)	
Grouping	By DeviceName within 15 Minutes	

KQL Query



Analytics rule details

Name: WannaCry File Hash or Dropped File Detected

Description: Detects WannaCry ransomware indicators on endpoints including known file hashes (diskpart.exe WannaCry binary), known WannaCry dropped filenames (@WanaDecryptor@.exe, taskdl.exe, taskse.exe), mass .WNCRY file creation indicating active encryption, and execution of system tool names from non-standard paths.

MITRE ATT&CK: Execution (1), Impact (1), Defense Evasion (1)

Severity: High

Status: Enabled

Analytics rule settings

Rule query:

```
let WannaCryHashes = dynamic([ "ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa", "84c82835a5d21bbcf75a61706d8ab549", "5ff465afaabcbf0150d1a3ab2c2e74f3a4426467", "4fef5e34143e646dbf9907c4374276f5" ]); let
WannaCryDroppedFiles = dynamic([ "@WanaDecryptor@.exe", "@Please_Read_Me@.txt", "taskdl.exe", "taskse.exe", "295021772622446.bat", "m.vbs", "u.wnry", "c.wnry" ]); let SystemToolsAbused = dynamic([ "diskpart.exe", "icacls.exe", "attrib.exe",
"cscript.exe" ]); // Detect known hashes executing let HashHits = DeviceProcessEvents | where TimeGenerated > ago(1h) | where SHA256 in (WannaCryHashes) or MD5 in (WannaCryHashes) | project TimeGenerated, DeviceName, AccountName, FileName,
FolderPath, SHA256, MD5, ProcessCommandLine, InitiatingProcessFileName | extend DetectionType = "KnownWannaCryHash" | extend AlertDetail = strcat("WannaCry hash detected: ", FileName, " on ", DeviceName); // Detect known dropped filenames
let DroppedFileHits = DeviceFileEvents | where TimeGenerated > ago(1h) | where FileName in (WannaCryDroppedFiles) or FileName endswith ".wnry" | project TimeGenerated, DeviceName, AccountName = InitiatingProcessAccountName, FileName,
FolderPath, SHA256, MD5, ProcessCommandLine = InitiatingProcessCommandLine, InitiatingProcessFileName | extend DetectionType = "WannaCryDroppedFile" | extend AlertDetail = strcat("WannaCry dropped file detected: ", FileName, " on ",
DeviceName); // Detect mass .WNCRY file creation (active encryption) let EncryptionHits = DeviceFileEvents | where TimeGenerated > ago(15m) | where FileName endswith ".WNCRY" | summarize EncryptedFileCount = count(), FirstSeen =
min(TimeGenerated), LastSeen = max(TimeGenerated) by DeviceName, InitiatingProcessFileName, InitiatingProcessAccountName | where EncryptedFileCount > 5 | extend TimeGenerated = FirstSeen | extend AccountName =
InitiatingProcessAccountName | extend FileName = strcat(tostring(EncryptedFileCount), " .WNCRY files created") | extend FolderPath = "Multiple directories" | extend SHA256 = "" | extend MD5 = "" | extend ProcessCommandLine =
InitiatingProcessFileName | extend DetectionType = "ActiveWannaCryEncryption" | extend AlertDetail = strcat("ACTIVE ENCRYPTION: ", tostring(EncryptedFileCount), " .WNCRY files on ", DeviceName, " - ISOLATE IMMEDIATELY") | project TimeGenerated,
DeviceName, AccountName, FileName, FolderPath, SHA256, MD5, ProcessCommandLine, InitiatingProcessFileName, DetectionType, AlertDetail; // Detect system tools running from non-standard paths let MasqueradingHits = DeviceProcessEvents | where
TimeGenerated > ago(1h) | where FileName in (SystemToolsAbused) | where FolderPath !startswith "C:\\Windows\\System32" and FolderPath !startswith "C:\\Windows\\SysWOW64" | project TimeGenerated, DeviceName, AccountName, FileName,
FolderPath, SHA256, MD5, ProcessCommandLine, InitiatingProcessFileName | extend DetectionType = "SystemToolMasquerading" | extend AlertDetail = strcat("System tool running from non-standard path: ", FileName, " in ", FolderPath, " on ",
DeviceName); // Combine all detection types union HashHits, DroppedFileHits, EncryptionHits, MasqueradingHits | order by TimeGenerated desc
```

RULE 4 - Sentinel Analytics rule saved, showing Critical severity and Enabled status

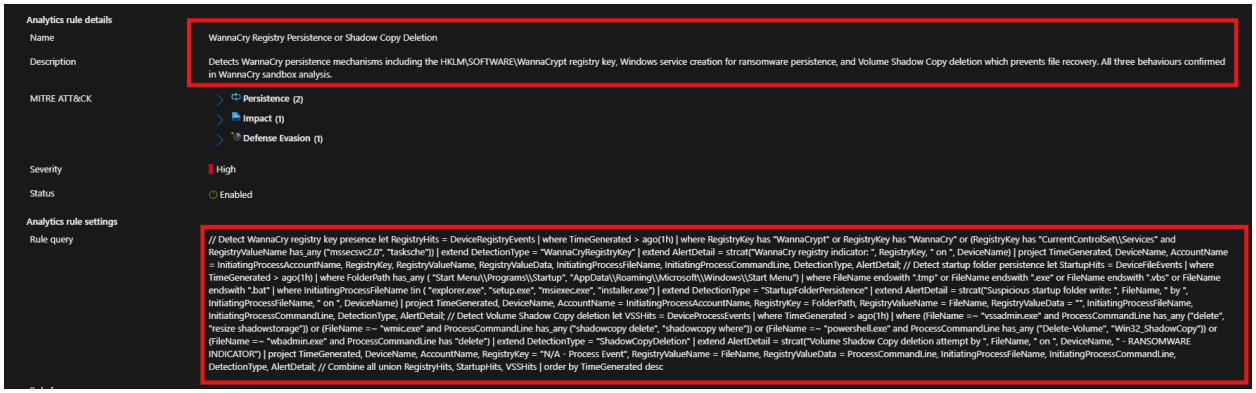
```
let WannaCryHashes = dynamic(["ed01ebfbc9eb5bbea545af4d01bf5f1071661840480439c6e5babe8e080e41aa",
"84c82835a5d21bbcf75a61706d8ab549", "5ff465afaabcbf0150d1a3ab2c2e74f3a4426467", "4fef5e34143e646dbf9907c4374276f5"]);
let WannaCryDroppedFiles = dynamic(["@WanaDecryptor@.exe", "@Please_Read_Me@.txt", "taskdl.exe", "taskse.exe", "295021772622446.bat",
"m.vbs", "u.wnry", "c.wnry"]);
DeviceProcessEvents
| where TimeGenerated > ago(15m)
| where SHA256 in (WannaCryHashes) or MD5 in (WannaCryHashes) or FileName in (WannaCryDroppedFiles)
| project TimeGenerated, DeviceName, AccountName, FileName, FolderPath, SHA256, MD5, ProcessCommandLine, InitiatingProcessFileName
| extend DetectionType = "WannaCryFileOrHash", AlertDetail = strcat("WannaCry indicator: ", FileName, " on ", DeviceName)
union (DeviceFileEvents | where TimeGenerated > ago(15m) | where FileName endswith ".WNCRY" | summarize
EncryptedFileCount=count(), FirstSeen=min(TimeGenerated) by DeviceName, InitiatingProcessFileName, InitiatingProcessAccountName |
where EncryptedFileCount > 5 | extend TimeGenerated=FirstSeen, AccountName=InitiatingProcessAccountName,
FileName=strcat(tostring(EncryptedFileCount), " .WNCRY files created"), FolderPath="Multiple directories",
DetectionType="ActiveWannaCryEncryption", AlertDetail=strcat("ACTIVE ENCRYPTION: ", tostring(EncryptedFileCount), " .WNCRY files
on ", DeviceName, " - ISOLATE IMMEDIATELY"))
```

RULE 4 - KQL

RULE 5	WannaCry Registry Persistence / Shadow Copy Deletion	CRITICAL

Field	Value
Description	Detects WannaCry persistence via HKLM\SOFTWARE\WannaCrypt registry key and known service names, startup folder writes by unexpected processes, and Volume Shadow Copy deletion.
MITRE Tactic	Persistence | Impact | Defense Evasion
MITRE Technique	T1547.001 - Registry Run Keys | T1490 - Inhibit System Recovery | T1070 - Indicator Removal
Run Frequency	Every 15 Minutes
Lookback Window	15 Minutes
Alert Threshold	Greater than 0 results
Entity Mapping	Host (DeviceName), Account (AccountName), RegistryKey (RegistryKey)
Grouping	By DeviceName within 15 Minutes

KQL Query



Analytics rule details

Name: WannaCry Registry Persistence or Shadow Copy Deletion

Description: Detects WannaCry persistence mechanisms including the HKLM\SOFTWARE\WannaCrypt registry key, Windows service creation for ransomware persistence, and Volume Shadow Copy deletion which prevents file recovery. All three behaviours confirmed in WannaCry sandbox analysis.

MITRE ATT&CK: Persistence (2); Impact (1); Defense Evasion (1)

Severity: High

Status: Enabled

Analytics rule settings

Rule query:

```
// Detect WannaCry registry key presence let RegistryHits = DeviceRegistryEvents | where TimeGenerated > ago(1h) | where RegistryKey has "WannaCrypt" or RegistryKey has "WannaCry" or (RegistryKey has "CurrentControlSet\\Services" and
RegistryValueName has_any ("mssecsvc2.0", "tasksche")) | extend DetectionType = "WannaCryRegistryKey" | extend AlertDetail = strcat("WannaCry registry indicator: ", RegistryKey, " on ", DeviceName) | project TimeGenerated, DeviceName, AccountName
= InitiatingProcessAccountName, RegistryKey, RegistryValueName, RegistryValueData, InitiatingProcessFileName, InitiatingProcessCommandLine, DetectionType, AlertDetail; // Detect startup folder persistence let StartupHits = DeviceFileEvents | where
TimeGenerated > ago(1h) | where FolderPath has_any ("Start Menu\\Programs\\Startup", "AppData\\Roaming\\Microsoft\\Windows\\Start Menu") | where FileName endswith ".tmp" or FileName endswith ".exe" or FileName endswith ".vbs" or FileName
endswith ".bat" | where InitiatingProcessFileName !in ("explorer.exe", "setup.exe", "msiexec.exe", "installer.exe") | extend DetectionType = "StartupFolderPersistence" | extend AlertDetail = strcat("Suspicious startup folder write: ", FileName, " by ",
InitiatingProcessFileName, " on ", DeviceName) | project TimeGenerated, DeviceName, AccountName = InitiatingProcessAccountName, RegistryKey = FolderPath, RegistryValueName = FileName, RegistryValueData = "", InitiatingProcessFileName,
InitiatingProcessCommandLine, DetectionType, AlertDetail; // Detect Volume Shadow Copy deletion let VSSHits = DeviceProcessEvents | where TimeGenerated > ago(1h) | where (FileName =~ "vssadmin.exe" and ProcessCommandLine has_any ("delete",
"resize shadowstorage")) or (FileName =~ "wmic.exe" and ProcessCommandLine has_any ("shadowcopy delete", "shadowcopy where")) or (FileName =~ "powershell.exe" and ProcessCommandLine has_any ("Delete-Volume", "Win32_ShadowCopy")) or
(FileName =~ "wbadmin.exe" and ProcessCommandLine has "delete") | extend DetectionType = "ShadowCopyDeletion" | extend AlertDetail = strcat("Volume Shadow Copy deletion attempt by ", FileName, " on ", DeviceName, " - RANSOMWARE
INDICATOR") | project TimeGenerated, DeviceName, AccountName, RegistryKey = "N/A - Process Event", RegistryValueName = FileName, RegistryValueData = ProcessCommandLine, InitiatingProcessFileName, InitiatingProcessCommandLine,
DetectionType, AlertDetail; // Combine all union RegistryHits, StartupHits, VSSHits | order by TimeGenerated desc
```

RULE 5 - Sentinel Analytics rule saved, showing Critical severity and Enabled status

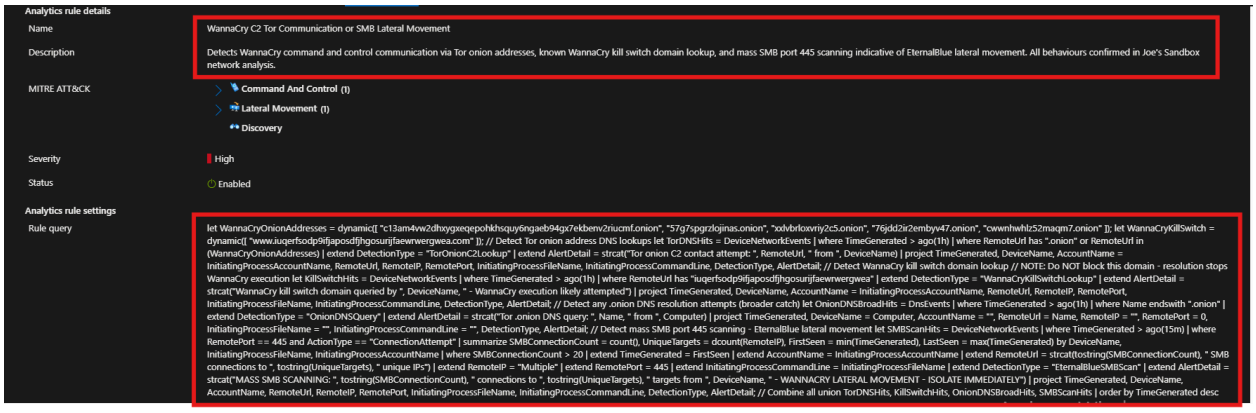
```
let RegistryHits = DeviceRegistryEvents
| where TimeGenerated > ago(15m)
| where RegistryKey has "WannaCrypt" or RegistryKey has "WannaCry" or (RegistryKey has "CurrentControlSet\Services" and
RegistryValueName has_any ("mssecsvc2.0", "tasksche"))
| extend DetectionType="WannaCryRegistryKey", AlertDetail=strcat("WannaCry registry: ", RegistryKey, " on ", DeviceName);
let VSSHits = DeviceProcessEvents
| where TimeGenerated > ago(15m)
| where (FileName =~ "vssadmin.exe" and ProcessCommandLine has_any ("delete", "resize shadowstorage")) or (FileName =~ "wmic.exe" and
ProcessCommandLine has_any ("shadowcopy delete")) or (FileName =~ "powershell.exe" and ProcessCommandLine has_any ("Delete-Volume",
"Win32_ShadowCopy")) or (FileName =~ "wbadmin.exe" and ProcessCommandLine has "delete")
| extend DetectionType="ShadowCopyDeletion", AlertDetail=strcat("VSS deletion by ", FileName, " on ", DeviceName, " - RANSOMWARE
INDICATOR");
union RegistryHits, VSSHits | order by TimeGenerated desc
```

RULE 5 - KQL

RULE 6	WannaCry C2 / Tor Communication / SMB Lateral Movement	CRITICAL

Field	Value
Description	Detects WannaCry C2 communication via Tor onion addresses, kill switch domain lookup, and mass SMB port 445 scanning indicative of EternalBlue lateral movement.
MITRE Tactic	Command and Control | Lateral Movement | Discovery
MITRE Technique	T1090 - Proxy | T1021 - Remote Services | T1046 - Network Service Discovery
Run Frequency	Every 15 Minutes
Lookback Window	15 Minutes
Alert Threshold	Greater than 0 results
Entity Mapping	Host (DeviceName), IP (RemoteIP), URL (RemoteUrl)
Grouping	By DeviceName within 15 Minutes

KQL Query



Analytics rule details

Name: WannaCry C2 Tor Communication or SMB Lateral Movement

Description: Detects WannaCry command and control communication via Tor onion addresses, known WannaCry kill switch domain lookup, and mass SMB port 445 scanning indicative of EternalBlue lateral movement. All behaviours confirmed in Joe's Sandbox network analysis.

MITRE ATT&CK: Command And Control (1); Lateral Movement (1); Discovery

Severity: High

Status: Enabled

Analytics rule settings

Rule query:

```
let WannaCryOnionAddresses = dynamic([ "c13am4vw2dhxygxeqepohkhsquy6ngaeb94gx7ekbenv2riucmf.onion", "57g7spgrzlojinas.onion", "xxlvbrloxvriy2c5.onion", "76jdd2ir2embyv47.onion", "owwnhwhlz52maqm7.onion" ]); let WannaCryKillSwitch =
dynamic([ "www.iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea.com" ]); // Detect Tor onion address DNS lookups let TorDNSHits = DeviceNetworkEvents | where TimeGenerated > ago(1h) | where RemoteUrl has ".onion" or RemoteUrl in
(WannaCryOnionAddresses) | extend DetectionType = "TorOnionC2Lookup" | extend AlertDetail = strcat("Tor onion C2 contact attempt: ", RemoteUrl, " from ", DeviceName) | project TimeGenerated, DeviceName, AccountName =
InitiatingProcessAccountName, RemoteUrl, RemoteIP, RemotePort, InitiatingProcessFileName, InitiatingProcessCommandLine, DetectionType, AlertDetail; // Detect WannaCry kill switch domain lookup // NOTE: Do NOT block this domain - resolution stops
WannaCry execution let KillSwitchHits = DeviceNetworkEvents | where TimeGenerated > ago(1h) | where RemoteUrl has "iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea" | extend DetectionType = "WannaCryKillSwitchLookup" | extend AlertDetail =
strcat("WannaCry kill switch domain queried by ", DeviceName, " - WannaCry execution likely attempted") | project TimeGenerated, DeviceName, AccountName = InitiatingProcessAccountName, RemoteUrl, RemoteIP, RemotePort,
InitiatingProcessFileName, InitiatingProcessCommandLine, DetectionType, AlertDetail; // Detect any .onion DNS resolution attempts (broader catch) let OnionDNSBroadHits = DnsEvents | where TimeGenerated > ago(1h) | where Name endswith ".onion" |
extend DetectionType = "OnionDNSQuery" | extend AlertDetail = strcat("Tor .onion DNS query: ", Name, " from ", Computer) | project TimeGenerated, DeviceName = Computer, AccountName = "", RemoteUrl = Name, RemoteIP = "", RemotePort = 0,
InitiatingProcessFileName = "", InitiatingProcessCommandLine = "", DetectionType, AlertDetail; // Detect mass SMB port 445 scanning - EternalBlue lateral movement let SMBScanHits = DeviceNetworkEvents | where TimeGenerated > ago(15m) | where
RemotePort == 445 and ActionType == "ConnectionAttempt" | summarize SMBConnectionCount = count(), UniqueTargets = dcount(RemoteIP), FirstSeen = min(TimeGenerated), LastSeen = max(TimeGenerated) by DeviceName,
InitiatingProcessFileName, InitiatingProcessAccountName | where SMBConnectionCount > 20 | extend TimeGenerated = FirstSeen | extend AccountName = InitiatingProcessAccountName | extend RemoteUrl = strcat(tostring(SMBConnectionCount), " SMB
connections to ", tostring(UniqueTargets), " unique IPs") | extend RemoteIP = "Multiple" | extend RemotePort = 445 | extend InitiatingProcessCommandLine = InitiatingProcessFileName | extend DetectionType = "EternalBlueSMBScan" | extend AlertDetail =
strcat("MASS SMB SCANNING: ", tostring(SMBConnectionCount), " connections to ", tostring(UniqueTargets), " targets from ", DeviceName, " - WANNACRY LATERAL MOVEMENT - ISOLATE IMMEDIATELY") | project TimeGenerated, DeviceName,
AccountName, RemoteUrl, RemoteIP, RemotePort, InitiatingProcessFileName, InitiatingProcessCommandLine, DetectionType, AlertDetail; // Combine all union TorDNSHits, KillSwitchHits, OnionDNSBroadHits, SMBScanHits | order by TimeGenerated desc
```

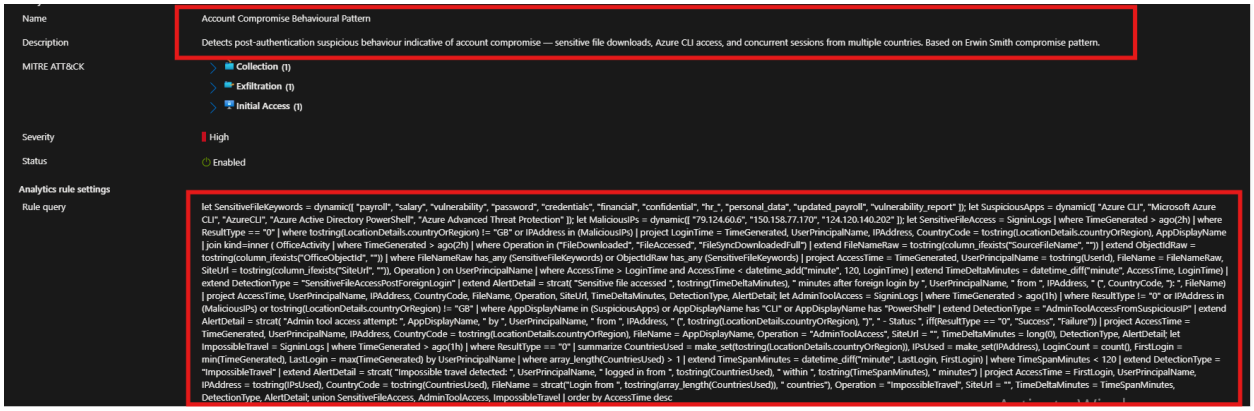
RULE 6 - Sentinel Analytics rule saved, showing Critical severity and Enabled status

```
let WannaCryOnion = dynamic(["c13am4vw2dhxygxeqepohkhsquy6ngaeb94gx7ekbenv2riucmf.onion", "57g7spgrzlojinas.onion",
"xxlvbrloxvriy2c5.onion", "76jdd2ir2embyv47.onion", "cwwnhwhlz52maqm7.onion"]);
let TorDNSHits = DeviceNetworkEvents | where TimeGenerated > ago(15m) | where RemoteUrl has ".onion" or RemoteUrl in
(WannaCryOnion) | extend DetectionType="TorOnionC2", AlertDetail=strcat("Tor C2: ", RemoteUrl, " from ", DeviceName);
let KillSwitchHits = DeviceNetworkEvents | where TimeGenerated > ago(15m) | where RemoteUrl has
"iuqerfsodp9ifjaposdfjhgosurijfaewrwergwea" | extend DetectionType="WannaCryKillSwitchLookup", AlertDetail=strcat("Kill switch domain
queried by ", DeviceName, " - WannaCry likely active");
let SMBScanHits = DeviceNetworkEvents | where TimeGenerated > ago(15m) | where RemotePort == 445 and ActionType ==
"ConnectionAttempt" | summarize SMBCount=count(), UniqueTargets=dcount(RemoteIP), FirstSeen=min(TimeGenerated) by DeviceName,
InitiatingProcessFileName, InitiatingProcessAccountName | where SMBCount > 20 | extend TimeGenerated=FirstSeen,
DetectionType="EternalBlueSMBScan", AlertDetail=strcat("MASS SMB SCAN: ", tostring(SMBCount), " connections from ", DeviceName, " -
ISOLATE IMMEDIATELY");
union TorDNSHits, KillSwitchHits, SMBScanHits | order by TimeGenerated desc
```

RULE 6 - KQL

RULE 7	Account Compromise Behavioural Pattern	CRITICAL
Field	**Value**	
Description	Detects post-authentication suspicious behaviour: sensitive file access after foreign login, admin tool access from suspicious IPs, and impossible travel. Based on the Erwin Smith compromise pattern in Task 3.	
MITRE Tactic	Collection | Exfiltration | Initial Access	
MITRE Technique	T1213 - Data from Information Repositories | T1048 - Exfiltration Over Alternative Protocol | T1078 - Valid Accounts	
Run Frequency	Every 15 Minutes	
Lookback Window	2 Hours	
Alert Threshold	Greater than 0 results	
Entity Mapping	Account (UserPrincipalName), IP (IPAddress), File (FileName)	
Grouping	By UserPrincipalName within 1 Hour	
Note	Uses column_ifexists() for OfficeActivity fields to handle schema variations across workspaces	

KQL Query



RULE 7 - Sentinel Analytics rule saved, showing Critical severity and Enabled status

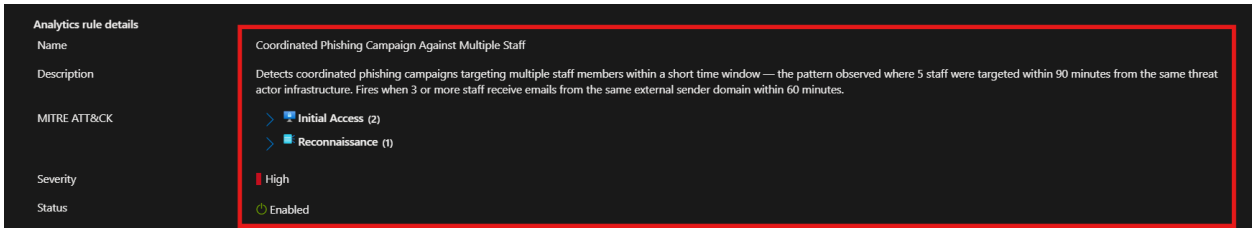
```
let SensitiveFileKeywords = dynamic(["payroll", "salary", "vulnerability", "password", "credentials", "financial", "confidential", "hr_",
"personal_data", "updated_payroll", "vulnerability_report"]);
let SuspiciousApps = dynamic(["Azure CLI", "Microsoft Azure CLI", "AzureCLI", "Azure Active Directory PowerShell"]);
let MaliciousIPs = dynamic(["79.124.60.6", "150.158.77.170", "124.120.140.202"]);
let SensitiveFileAccess = SigninLogs
| where TimeGenerated > ago(2h) and ResultType == "0"
| where tostring(LocationDetails.countryOrRegion) != "GB" or IPAddress in (MaliciousIPs)
| project LoginTime=TimeGenerated, UserPrincipalName, IPAddress, CountryCode=tostring(LocationDetails.countryOrRegion),
AppDisplayName
| join kind=inner (OfficeActivity | where TimeGenerated > ago(2h) | where Operation in ("FileDownloaded", "FileAccessed",
"FileSyncDownloadedFull") | extend FileNameRaw=tostring(column_ifexists("SourceFileName", "")),
ObjectIdRaw=tostring(column_ifexists("OfficeObjectId", "")) | where FileNameRaw has_any (SensitiveFileKeywords) or ObjectIdRaw has_any
(SensitiveFileKeywords) | project AccessTime=TimeGenerated, UserPrincipalName=tostring(UserId), FileName=FileNameRaw,
SiteUrl=tostring(column_ifexists("SiteUrl", "")), Operation) on UserPrincipalName
| where AccessTime > LoginTime and AccessTime < datetime_add("minute", 120, LoginTime)
| extend DetectionType="SensitiveFileAccessPostForeignLogin";
let AdminToolAccess = SigninLogs | where TimeGenerated > ago(1h) | where ResultType != "0" or IPAddress in (MaliciousIPs) or
tostring(LocationDetails.countryOrRegion) != "GB" | where AppDisplayName in (SuspiciousApps) or AppDisplayName has "CLI" or
AppDisplayName has "PowerShell" | extend DetectionType="AdminToolAccessFromSuspiciousIP";
union SensitiveFileAccess, AdminToolAccess | order by AccessTime desc
```

RULE 7 - KQL

RULE 8	Coordinated Phishing Campaign Against Multiple Staff	HIGH

Field	Value
Description	Detects coordinated phishing campaigns targeting multiple staff within a short window. Also detects display name spoofing of internal colleagues and new unknown senders targeting 3+ staff.
MITRE Tactic	Initial Access | Reconnaissance
MITRE Technique	T1566.001 - Spearphishing Attachment | T1598 - Phishing for Information
Run Frequency	Every 30 Minutes
Lookback Window	1 Hour
Alert Threshold	Greater than 0 results
Entity Mapping	Account (SenderFromAddress), Mailbox (Targets)
Grouping	By SenderFromDomain within 1 Hour
Note	Uses union isfuzzy=true with datatable fallback for IdentityInfo to handle workspaces where the table is not available

KQL Query



Analytics rule settings

Rule query

```
let PhishingDomains = dynamic([ "wtools.io", "kanzalshamsprojectmgt.com", "updetadobe.blogspot.com", "theannoyingsite.com", "azxmarket.com", "duchessgarden.sn" ]); let KnownSenders =
dynamic([ "alfredegov@gmail.com", "zelatcol@gmail.com", "projectdept@kanzalshamsprojectmgt.com", "c6d90d50-df87-11ee-a2fc-525400450766@azxmarket.com" ]); let InternalIdentities = union
isfuzzy=true ( IdentityInfo | where TimeGenerated > ago(7d) | project InternalDisplayName = tostring(AccountDisplayName), InternalUPN = tostring(AccountUPN) ), (
datatable(InternalDisplayName:string, InternalUPN:string)[] ); let MultiTargetDomain = EmailEvents | where TimeGenerated > ago(1h) | where SenderFromDomain in (PhishingDomains) or
SenderFromAddress in (KnownSenders) or SenderMailFromDomain in (PhishingDomains) | summarize TargetCount = dcount(RecipientEmailAddress), Targets = make_set(RecipientEmailAddress),
Subjects = make_set(Subject), FirstSeen = min(TimeGenerated), LastSeen = max(TimeGenerated) by SenderFromDomain, SenderFromAddress | where TargetCount >= 2 | extend CampaignDuration =
datetime_diff("minute", LastSeen, FirstSeen) | extend DetectionType = "KnownPhishingDomainMultiTarget" | extend AlertDetail = strcat( "Known phishing domain targeting ", tostring(TargetCount), "
staff: ", tostring(Targets), " from ", SenderFromAddress, " over ", tostring(CampaignDuration), " minutes") | project TimeGenerated = FirstSeen, SenderFromAddress, SenderFromDomain, TargetCount,
Targets = tostring(Targets), Subjects = tostring(Subjects), CampaignDuration, DetectionType, AlertDetail; let MultiTargetUnknown = EmailEvents | where TimeGenerated > ago(1h) | where
SenderFromDomain !in ( "company.com", "microsoft.com", "google.com", "outlook.com", "office365.com", "microsoftonline.com") | extend DeliveryActionSafe =
tostring(column_ifexists("DeliveryAction", "")) | extend AttachmentCountSafe = toint(column_ifexists("AttachmentCount", 0)) | where DeliveryActionSafe !in ("Blocked", "Quarantined") | summarize
TargetCount = dcount(RecipientEmailAddress), Targets = make_set(RecipientEmailAddress), Subjects = make_set(Subject), HasAttachment = max(iff(AttachmentCountSafe > 0, 1, 0)), FirstSeen =
min(TimeGenerated), LastSeen = max(TimeGenerated) by SenderFromDomain, SenderFromAddress | where TargetCount >= 3 | extend CampaignDuration = datetime_diff("minute", LastSeen,
FirstSeen) | where CampaignDuration <= 60 | extend DetectionType = iff(HasAttachment == 1, "NewCampaignWithAttachment", "NewCampaignNoAttachment") | extend AlertDetail = strcat(
"Potential new phishing campaign: ", tostring(TargetCount), " staff targeted by ", SenderFromAddress, " in ", tostring(CampaignDuration), " minutes. Attachment: ", iff(HasAttachment == 1, "YES",
"NO")) | project TimeGenerated = FirstSeen, SenderFromAddress, SenderFromDomain, TargetCount, Targets = tostring(Targets), Subjects = tostring(Subjects), CampaignDuration, DetectionType,
AlertDetail; let DisplayNameSpoof = EmailEvents | where TimeGenerated > ago(1h) | where SenderFromDomain !in ("company.com", "microsoft.com") | extend SenderDisplayNameSafe =
tostring(column_ifexists("SenderDisplayName", "")) | join kind=inner ( InternalIdentities ) on $left.SenderDisplayNameSafe == $right.InternalDisplayName | where SenderFromAddress !contains
"company.com" | extend DetectionType = "DisplayNameSpoofing" | extend AlertDetail = strcat( "Display name spoofing detected: External sender ", SenderFromAddress, " using internal staff name: ",
SenderDisplayNameSafe, " targeting ", RecipientEmailAddress) | project TimeGenerated, SenderFromAddress, SenderFromDomain, TargetCount = long(1), Targets = tostring(RecipientEmailAddress),
Subjects = tostring(Subject), CampaignDuration = long(0), DetectionType, AlertDetail; union MultiTargetDomain, MultiTargetUnknown, DisplayNameSpoof | order by TimeGenerated desc
```

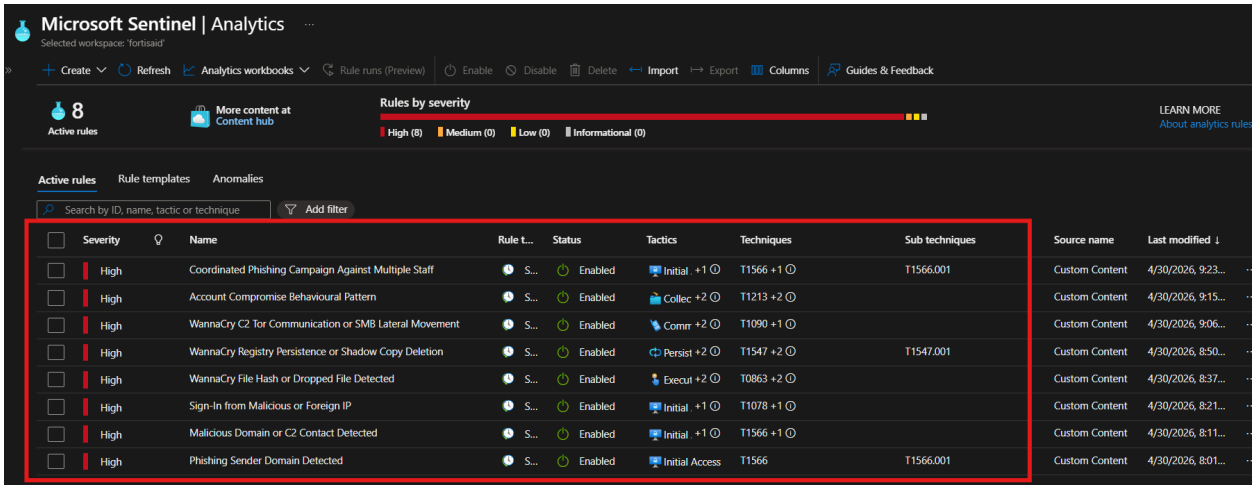
RULE 8 - Sentinel Analytics rule saved, showing High severity and Enabled status

```
let PhishingDomains = dynamic(["wtools.io", "kanzalshamsprojectmgt.com", "updetadobe.blogspot.com", "theannoyingsite.com",
"azxmarket.com", "duchessgarden.sn"]);
let KnownSenders = dynamic(["alfredegov@gmail.com", "zelatcol@gmail.com", "projectdept@kanzalshamsprojectmgt.com",
"c6d90d50-df87-11ee-a2fc-525400450766@azxmarket.com"]);
let InternalIdentities = union isfuzzy=true (IdentityInfo | where TimeGenerated > ago(7d) | project
InternalDisplayName=tostring(AccountDisplayName), InternalUPN=tostring(AccountUPN)), (datatable(InternalDisplayName:string,
InternalUPN:string)[]);
let MultiTargetDomain = EmailEvents | where TimeGenerated > ago(1h) | where SenderFromDomain in (PhishingDomains) or
SenderFromAddress in (KnownSenders) or SenderMailFromDomain in (PhishingDomains) | summarize
TargetCount=dcount(RecipientEmailAddress), Targets=make_set(RecipientEmailAddress), Subjects=make_set(Subject),
FirstSeen=min(TimeGenerated), LastSeen=max(TimeGenerated) by SenderFromDomain, SenderFromAddress | where TargetCount >= 2 |
extend DetectionType="KnownPhishingDomainMultiTarget";
let MultiTargetUnknown = EmailEvents | where TimeGenerated > ago(1h) | where SenderFromDomain !in ("company.com", "microsoft.com",
"google.com", "outlook.com", "office365.com", "microsoftonline.com") | extend DeliveryActionSafe=tostring(column_ifexists("DeliveryAction",
"")), AttachmentCountSafe=toint(column_ifexists("AttachmentCount", 0)) | where DeliveryActionSafe !in ("Blocked", "Quarantined") |
summarize TargetCount=dcount(RecipientEmailAddress), Targets=make_set(RecipientEmailAddress), Subjects=make_set(Subject),
HasAttachment=max(iff(AttachmentCountSafe>0,1,0)), FirstSeen=min(TimeGenerated), LastSeen=max(TimeGenerated) by
SenderFromDomain, SenderFromAddress | where TargetCount >= 3 | extend CampaignDuration=datetime_diff("minute", LastSeen, FirstSeen)
| where CampaignDuration <= 60 | extend DetectionType=iff(HasAttachment==1, "NewCampaignWithAttachment",
"NewCampaignNoAttachment");
let DisplayNameSpoof = EmailEvents | where TimeGenerated > ago(1h) | where SenderFromDomain !in ("company.com", "microsoft.com") |
extend SenderDisplayNameSafe=tostring(column_ifexists("SenderDisplayName", "")) | join kind=inner (InternalIdentities) on
$left.SenderDisplayNameSafe == $right.InternalDisplayName | where SenderFromAddress !contains "company.com" | extend
DetectionType="DisplayNameSpoofing";
union MultiTargetDomain, MultiTargetUnknown, DisplayNameSpoof | order by TimeGenerated desc
```

RULE 8 - KQL

5. All Rules - Sentinel Analytics View

The following screenshot shows all eight custom analytics rules as they appear in the Microsoft Sentinel Analytics blade, confirming successful creation and enabled status across all rules.



Microsoft Sentinel Analytics - all 8 rules visible showing Name, Severity, Status (Enabled), and Last Modified